John Ryan

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Summary

Senior Research Engineer with over 8 years of machine learning experience, from developing production NLP systems in 2016 to scaling frontier AI models for drug discovery at Isomorphic Labs. Specializes in performance optimization, distributed training, and accelerating inference. Brings a strong entrepreneurial background from founding deep-tech ventures, with proven ability to lead technical projects from research to production.

Experience

ISOMORPHIC LABS | Senior Research Engineer - Performance and Scaling

Jan 2026 - Present | London, UK

- **Executed core research** on frontier AI models, alongside developing advanced performance and scaling solutions to push the boundaries of computational efficiency.
- **Enabled** compatibility for core models on a new class of hardware, unlocking large-scale inference on thousands of accelerators and enhancing computational flexibility.
- **Scaled** single training jobs to run efficiently on high-hundreds of accelerators by resolving complex multi-node communication bottlenecks/issues.

| Research Engineer II - Applied ML Research

Nov 2024 - Dec 2025

- **Directed** the technical strategy and roadmap for the inference domain within the Applied Research team, establishing best practices and guiding the direction of key engineering efforts.
- **Architected and spearheaded** the development of a scalable internal platform for repeatable ML inference workflows for repeatable science, now used by 4+ teams.
- **Engineered** a deep optimization initiative for core inference pipelines, reducing end-to-end latency by over 98% (from >10 minutes to <10 seconds) through custom kernel development and targeted profiling.
- **Technical contributor** to the proprietary IsoDDE for end-to-end drug discovery, achieving state-of-the-art performance in protein-ligand structure prediction and binding affinity.

| Research Engineer - Applied ML Research

Feb 2024 - Oct 2024

- **Led** machine learning for two drug discovery programs, establishing a robust feedback loop between AI prediction and experimental validation to prioritize high-potential therapeutic candidates.
- **Onboarded and mentored** several new engineers, providing technical guidance to accelerate their contributions to core projects and their transition to drug discovery.
- **Performed and implemented research** on various topics, including experimental conditioning, evaluation protocols for novel tasks, inference time guidance and improving model computational performance.

ENTREPRENEUR FIRST | Founder in Residence

Oct 2023 - Jan 2024 | London, UK

- Selected for the globally competitive talent investor program to conceptualize and build a deep-tech venture from the ground up.
- Architected and developed rapid technical prototypes across multiple AI domains, including LLM-based computer control agents, Reinforcement Learning (RL) as a Service, and ML interpretability tools.

DUCTILE AI | Founder

July 2022 - Sep 2023 | London, UK

- **Engineered** robust Visual SLAM systems for resource-constrained aerial drones; rapidly advanced from research to functional prototypes deployed for field testing in Ukraine and Estonia.
- **Drove** the research strategy, evaluating and implementing novel approaches combining Neural Radiance Fields (NeRFs) and Bayesian Modelling.
- **Architected** and optimized the full MLOps pipeline using ROS, demonstrating deep technical leadership in a production-focused environment.

RIGR AI | Research Lead

June 2022 - Aug 2023 | London, UK (Contract)

- **Directed** multiple concurrent R&D projects in NLP, Computer Vision, and Reinforcement Learning for external clients.

- ➔ **Owned** the deployment of research models into production, focusing on scalability and performance optimization.
- ➔ **Served** as a key technical partner in the €2M EU-funded ARICA consortium, developing the Video Summarisation Tool (VST) to help law enforcement investigate and prioritize high-risk cases on the darknet.

LINKEDIN | Machine Learning Engineering Intern

Summers 2019, 2020, 2021 | Dublin, Ireland

- ➔ **Developed and productionized** active learning and few-shot learning systems for NLP using Bayesian ML, PyTorch, and internal distributed training frameworks.
- ➔ **Architected** a scalable ML pipeline for large-scale personalized generative text models using Spark, Tensorflow, and BERT.
- ➔ **Built** and delivered the core tooling for machine learning interpretability to debug models and datasets, integrated with Tensorflow and Tensorboard.

TEAMWORK.COM | Software Developer

Summers 2017 & 2018 | Cork, Ireland

- ➔ Engineered and shipped new backend features for the Teamwork Desk platform, and explored early-stage applications of NLP to enhance product functionality. (2018)
- ➔ Built and benchmarked a proof-of-concept text similarity search engine using Lucene and text embeddings to evaluate its potential for new product features. (2017)

INTEL SECURITY | Data Science Intern

Summer 2016 | Cork, Ireland

- ➔ Developed production NLP systems using Gradient Boosted Trees & CNNs; this work directly led to co-authoring and presenting an academic paper on online harassment detection in 2017.

Publications & Research

ISOMORPHIC LABS DRUG DESIGN ENGINE (ISODDE) | Technical Report | 2026

- ➔ Served as a technical contributor to the proprietary IsoDDE for end-to-end drug discovery, achieving state-of-the-art performance in protein-ligand structure prediction and binding affinity.

REINFORCEMENT LEARNING FOR SPACE OPERATIONS | MSc Thesis | 2022

- ➔ Designed and implemented a fully differentiable simulator for satellite operations on GPUs to enable gradient-based learning.
- ➔ Developed and benchmarked safe reinforcement learning policies for satellite control in collaboration with the European Space Agency (ESA).

BATCH ACTIVE LEARNING WITH DEEP KERNEL GAUSSIAN PROCESS CLASSIFIERS | Undergraduate Thesis (Distinction) | 2021

- ➔ Implemented and benchmarked Bayesian Active Learning methods with Deep Gaussian Process models in PyTorch.
- ➔ Devised and proved a method to improve the computational complexity of batch selection, making it viable for larger datasets.

HACK HARASSMENT: TECHNOLOGY SOLUTIONS TO COMBAT ONLINE HARASSMENT | ALW1: 1st Workshop on Abusive Language Online | 2017

- ➔ Co-authored and presented work on production NLP systems using Gradient Boosted Trees and CNNs for harassment detection.

Education

UNIVERSITY OF OXFORD Masters - Computer Science - Merit

Oct 2021 - June 2022

Bachelors - Computer Science - First Class Honours

Oct 2018- June 2021

Selected Awards: 3x IOI National Delegate (2016-2018) | 1st Place IrICPC (2018) | MLH Hackathon Winner

Skills

Languages: Python, C++, CUDA, Golang, Javascript, Julia, Java, Scala, Bash
AI Frameworks & Libraries: PyTorch, JAX, XLA, Mosaic, Triton, Pallas, Tensorflow, Keras
Distributed Systems & MLOps: Docker, Kubernetes, DDP, FSDP, ROS, Linux, Spark

Honors & Awards

SELECTED AWARDS

- Winner - OxfordHack @ Major League Hacking (2018)
- 1st Place Overall & 1st Place Junior - Irish Collegiate Programming Competition (IrlCPC) (2018)
- Irish National Delegate - International Olympiad in Informatics (IOI), Japan (2018) & Iran (2017) & Russia (2016)